Retail Sales Optimizer

Upload your dataset → column auto-detection → variables creation → segmentation → discount optimization → categories recommendation

1) Dataset

Upload your CSV

retail_price.csv
121.4KB

+

☑ Use example dataset if no other csv is uploaded

2) Segmentation

More segments = more detail, but can be less stable.

Min segments
3

Max segments
10

Compression level (PCA)
0.90

☑ Balance segments (similar sizes)

ONLY balance the final assignation (do not change the model), to avoid tiny clusters.

1) Dataset

Fuente: **CSV uploaded** | Filas: **676** | Columnas: **30**

	product_id	product_category_name	month_year	qty	total_price	freight_price	unit_price	product_name_lenght	product_description_lenght	product_photos_qty	product_weight_g	product_score
0	bed1	bed_bath_table	01-05-2017	1	45.95	15.1	45.95	39	161	2	350	4
1	bed1	bed_bath_table	01-06-2017	3	137.85	12.9333	45.95	39	161	2	350	4
2	bed1	bed_bath_table	01-07-2017	6	275.7	14.84	45.95	39	161	2	350	4
3	bed1	bed_bath_table	01-08-2017	4	183.8	14.2875	45.95	39	161	2	350	4
4	bed1	bed_bath_table	01-09-2017	2	91.9	15.1	45.95	39	161	2	350	4
5	bed1	bed_bath_table	01-10-2017	3	137.85	15.1	45.95	39	161	2	350	4
6	bed1	bed_bath_table	01-11-2017	11	445.85	15.8327	40.5318	39	161	2	350	4
7	bed1	bed_bath_table	01-12-2017	6	239.94	15.23	39.99	39	161	2	350	4
8	bed1	bed_bath_table	01-01-2018	19	759.81	16.5337	39.99	39	161	2	350	4
9	bed1	bed_bath_table	01-02-2018	18	719.82	13.7494	39.99	39	161	2	350	4

› Warnings (per missing columns / inferences)

› Mapping detected (columna original → correct format)

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24	garden5	garden_tools	01-11-2017	17	1683	32.8635	99	36	450	1	9000	4.1

⋮

⌄ Warnings (per missing columns / inferences)

No order_id column; artificial column has been created.

No date columns; some temporal features will be 0/NaN.

There was no discount; synthetic discount_percent was generated(solo demo).

⌄ Mapping detected (columna original → correct format)

```
{
  "customer_id" : "customers"
  "product_category" : "product_category_name"
  "unit_price" : "unit_price"
  "quantity" : "qty"
  "total_amount" : "total_price"
}
```

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Variables created (preview)

	customer_id	product_category	unit_price	quantity	total_amount	discount_percent	discount_is_synthetic	order_prediscount	net_revenue	value_lost_discount	engagement_score	high_engagement_fla
4	54	bed_bath_table	45.95	2	91.9	0.0159	1	91.9	90.4386	1.4614	0	
5	50	bed_bath_table	45.95	3	137.85	0.3016	1	137.85	96.2754	41.5746	0	
6	97	bed_bath_table	40.5318	11	445.85	0.0497	1	445.85	423.6827	22.1673	0	
7	41	bed_bath_table	39.99	6	239.94	0	1	239.94	239.94	0	0	
8	62	bed_bath_table	39.99	19	759.81	0.0058	1	759.81	755.389	4.421	0	
9	62	bed_bath_table	39.99	18	719.82	0.0048	1	719.82	716.3893	3.4307	0	
10	43	bed_bath_table	39.99	17	679.83	0.0001	1	679.83	679.7302	0.0998	0	
11	34	bed_bath_table	39.99	13	519.87	0	1	519.87	519.87	0	0	
12	57	bed_bath_table	39.99	19	759.81	0.2513	1	759.81	568.8703	190.9397	0	
13	43	bed_bath_table	39.99	5	199.95	0.0493	1	199.95	190.0921	9.8579	0	

⚠ There was no discount; synthetic discount_percent was generated(solo demo)

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ONLY balance the final assignation (do not change the model), to avoid tiny clusters.

3) Optimal Discounts

Discounts to test (%)
0 × 5 × 10 × 15 × 20 × 25 ×

Minimun rows per segment to estimate
120

Anti-feeze (recommended if K is increased manually):

Max rows per segment (simulation)
4000

Max rows to train global demand
50000

4) Recomendations (ALS)

N° of recommended categories
5

Recommendator quality (factors)
96

2) Segmentation


```
{
  "k" : 4
  "silhouette" : 0.33392455914170227
  "min_segment_prop" : 0.10638297872340426
  "balance_penalty" : 1
}
```

Summary per segment (to interprete who's who):

	segment	clientes	gasto_total	recencia_dias	descuento_medio	engagement	entrega_media	diversidad
0	0	24	2276.6053	None	0.1352	0	0	1.625
1	1	24	8111.98	None	0.0993	0	0	2.0833
2	2	23	9251.5275	None	0.1055	0	0	1.0435
3	3	23	17453.7272	None	0.093	0	0	1.2174

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3) Optimal discount per segment

Calculate optimal discounts

Tip: adjust segmentation and then press the button. No recalculations when moving sliders.

Press the button to generate the policy.

4) Category Recommendator(ALS)

Train recommendator

Only train when pressing (avoiding constant recalculation).

Press 'Train recommendator' to activate recommendations.

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	segment	n_orders_segment_x	expected_avg_revenue_per_order	n_orders_segment_y	method	estado	Descuento recomendado (%)
0	0	93	None	None	None	Not calculated (not enough data in the segment)	None
1	1	235	1031.6795	235	synthetic_elasticity_prior	OK	5
2	2	168	1762.964	168	synthetic_elasticity_prior	OK	0
3	3	180	2564.6027	180	synthetic_elasticity_prior	OK	0

If a segment appears as 'Not calculated', it is because there are not enough data in the segment. Lower K or decrease minimun per segment.

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Choose a Customer_ID

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Segment of Client: 1

Discount recomendated for this segment: **5.0%**

Category Recommendator

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Category Recommendator

	Recommended category	Score	Afinidad (%)
0	garden_tools	8	100
1	bed_bath_table	0.076	1

Score/Afinidad = estimated preference of the client by caegory, based in historical behavior.